

Plasmon-enhanced photocatalytic hydrogen production on Au/TiO₂ hybrid nanocrystal arrays

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Highlights

- Highly regular hexagonal close-packed core-shell Au/TiO₂ hybrid nanocrystal arrays.
- Outstanding photocatalytic properties for methanol aqueous solution splitting.
- Exceptional improvement from strong coupling of the enhanced electric field.
- FDTD simulations demonstrating lateral coupling of plasmonically enhanced field.
- Optimal structures for maximum enhancement can be designed.

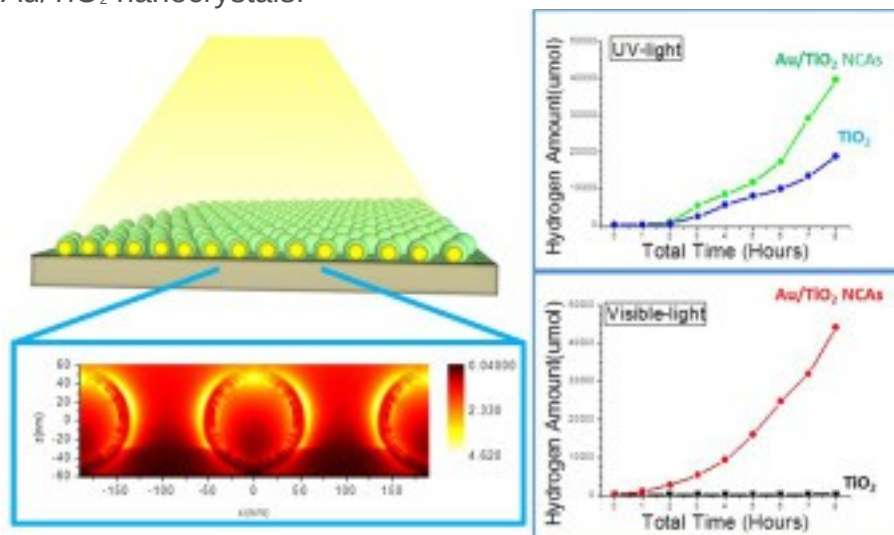
Abstract

In recent years, hybrid [nanostructure](#) consisted of plasmonic metals and different [dielectrics](#) have attracted much attention for their intriguing plasmonic properties. Recent studies have also shown that by introducing plasmonic metals, the photocatalytic efficiency of [semiconductor](#) can improve via plasmon-enhanced light absorption. In this work, we have demonstrated excellent photocatalytic properties for hydrogen production using hexagonal close-packed core-shell Au/TiO₂ hybrid [nanocrystal](#) arrays. Under both ultra-violet and visible light, significant increase in the hydrogen production from methanol (20%) solution splitting was achieved with the hybrid Au/TiO₂ nanocrystal arrays in comparison with bare TiO₂ thin film as well as randomly distributed Au/TiO₂ nanocrystals. From the [finite difference](#) time domain (FDTD) simulation, the significant increase (up to 60%) in hydrogen production

can be correlated to strong and optimum coupling of the enhanced electric field from localized surface plasmon resonance in Au/TiO₂ nanocrystal arrays. In addition to allowing more accurate measurement of plasmonic enhancement, the ordered nanostructures have been shown to be especially amenable to the systematic analysis of lateral coupling of plasmonically enhanced electric field. As a result, optimal structures with appropriate spacing of core-shell metal-dielectric nanocrystals, metal core size and dielectric shell thickness for maximum enhancement can be designed.

Graphical abstract

Under both ultra-violet and visible light, significant increase in the hydrogen production from methanol (20%) solution splitting was achieved with the hybrid Au/TiO₂ nanocrystal arrays in comparison with bare TiO₂ thin film as well as randomly distributed Au/TiO₂ nanocrystals.



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Keywords

Localized surface plasmon resonance

Metal nanocrystal array

Photocatalytic reactions

Hydrogen production

1. Introduction

Nowadays, various metal and [semiconductor materials](#) with intriguing plasmonic optical properties attracted much attention. The tremendous progresses in precise control of the shape, size, morphology and distribution of [nanocrystals](#) have made those localized surface plasmon resonance (LSPR) properties more amenable for analysis and fundamental understanding [1], [2], [3]. In particular, a previous work showed that the LSPR wavelengths of the Au nanocrystal arrays (NCAs) could be systematically tuned from the visible to near-infrared region by controlling the diameter of the Au nanocrystals and immersion of Au nanocrystals in silicon [2], [4].

However, to meet the requirements for various applications, hybrid [nanostructure](#) prepared by integrating plasmonic metal with different [dielectrics](#), such as SiO_2 , [ZnO](#), Fe_3O_4 , are of importance [4], [5], [6]. Compared with single-component plasmonic metal, the hybrid nanostructures exhibit extraordinary optical, electronic, and magnetic properties combining with unique LSPR properties [7], [8]. As the ubiquitous basic materials of modern electronic devices, [semiconducting materials](#) are widely investigated. On the other hand, a number of important [semiconductors](#) are often wide-bandgap materials and interact weakly with visible-light. Thus, in order to increase the response to visible-light of the semiconductor, many bandgap engineering methods, including doping C, N, or metal elements, have been explored [9], [10]. An alternative scheme, introducing plasmonic metal to semiconductor to form hybrid metal-semiconductor nanostructure has been considered a more promising approach [11], [12], [13], [14], [15]. The increase of visible-light absorption of the hybrid nanostructure can be attributed to two origins. First, under the visible-light resonating with the LSPR of the plasmonic metal, the localized electric field near the metal/semiconductor interface is enhanced by LSPR effect, hence increasing electron-hole pairs in the semiconductor region. Second, metal component may help separate the electron-hole pairs and increase [surface reactions](#).

Nevertheless, for the wavelength of the incident light with energy less than the bandgap of semiconductor, normally electron-hole pairs are not generated in the semiconductor. Once the incident light stimulated LSPR effect, the [electron-transfer](#) may occur in the hybrid nanostructure due to the [hot electron](#) effects [16], [17], [18]. Direct photoexcitation and plasmon decay are the two main possible hot electrons generation mechanisms. For plasmon-induced hot electrons, the transition was driven by the localized plasmon-induced near field [19], [20], [21]. The plasmonic hot electron generation is proportional to plasmon-induced near field intensity with enhancement, while occurring independently of bulk absorption [22], [23]. The plasmon-induced hot electrons generation would be efficient since the plasmonic absorbers can lead to near-

perfect absorption [24], [25]. Moreover, hot electrons generated by plasmon decay have higher energies than those by direct excitation, making plasmon-enhanced [photocatalysis](#) more efficient [26], [27].

In the present work, the Au/TiO₂ hybrid nanocrystal arrays were successfully fabricated. Combining colloidal [lithography](#), dewetting process driven by [surface energy](#) and atomic layer deposition (ALD), a large area of hexagonal close-packed Au/TiO₂ hybrid nanocrystal arrays with 100–110 nm single [crystalline](#) Au core, 10–40 nm TiO₂ shell are prepared with highly ordered periodicity. The spacing of nanocrystal arrays can be controlled by proper selection of colloidal size, while the Au core and TiO₂ shell by the thicknesses of deposition. Through tuning the thickness of TiO₂ of the nanocrystal arrays, the LSPR wavelength can be adjusted to red shift and splitting into two LSPR peaks. The scattering spectra of the nanocrystal arrays can be interpreted by the simulated results based on Mie theory.

The hybrid Au/TiO₂ nanocrystal arrays were employed as [photocatalysts](#) in aqueous methanol solution (20%) and compared with the bare TiO₂ thin film. Under both UV and visible light, the hydrogen production efficiency of the hybrid Au/TiO₂ nanocrystal arrays exhibited significant increase in the aqueous methanol solution splitting. From the [finite difference](#) time domain (FDTD) simulation, it is seen that the electric field in TiO₂ has been significantly enhanced by LSPR, confirming that the generation of electron and hole pairs was increased by plasmon-enhanced light absorption.

In addition, the regular spacing of NCAs affected the efficiency of hydrogen production compared to the randomly distributed Au/TiO₂ particles. The significant lateral electric field coupling in periodical NCAs may reinforce the electric field constructively, while the lateral coupling has been impaired with random core-to-core spacings. Furthermore, the near-field enhancement was found to increase with the size of Au core, which led to stronger lateral electric field coupling as well as higher photocatalytic efficiency.

2. Experimental

2.1. Fabrication of the hexagonal Au/TiO₂ hybrid nanocrystal arrays

The hexagonal close-packed Au/TiO₂ hybrid [nanocrystal](#) arrays were fabricated in two steps: the fabrication of hexagonal Au nanocrystal arrays and the deposition of TiO₂ by atomic layer deposition.

For the formation of Au nanocrystal arrays, the monolayer [polystyrene](#) spheres (PS spheres) were spread on the [sodium dodecyl sulfate](#) decorated glass substrates, then immersed into deionized water gradually. The monolayer was scooped up with a clean Si substrate. With appropriate power and time, the size of the PS spheres was

controlled with O₂ [plasma etching](#). The SiO₂ layer of different thickness was then deposited at a rate of 0.05 nm/s on the monolayer PS spheres arrays with an electron beam evaporation system under 10⁻⁶ Torr. To remove PS spheres and obtain the SiO₂ honeycomb cells arrays, samples were soaked in toluene with ultrasonic vibration for 5 h. Subsequently, the Au film was deposited on SiO₂ honeycomb cell arrays by electron beam deposition with various thicknesses at a rate of 0.1 nm/s under a vacuum of 10⁻⁶ Torr. The as-deposited samples were annealed in a [quartz](#) tube furnace at 1000 °C for 10 min in Ar atmosphere. The [metal films](#) were dewetted and formed hexagonal Au nanocrystal arrays..

After the colloidal [lithography](#) and the [annealing](#) process, TiO₂ layers of different thicknesses were deposited by atomic layer deposition. With tetrakis-dimethylamino [titanium](#) (TDMAT) as the precursor, the deposition cycle was set as 1 s/5 s/3 s/5 s at 0.057 nm/cycle. The hexagonal Au/TiO₂ hybrid nanocrystal arrays were then successfully fabricated.

2.2. Measurements of the scattering spectra

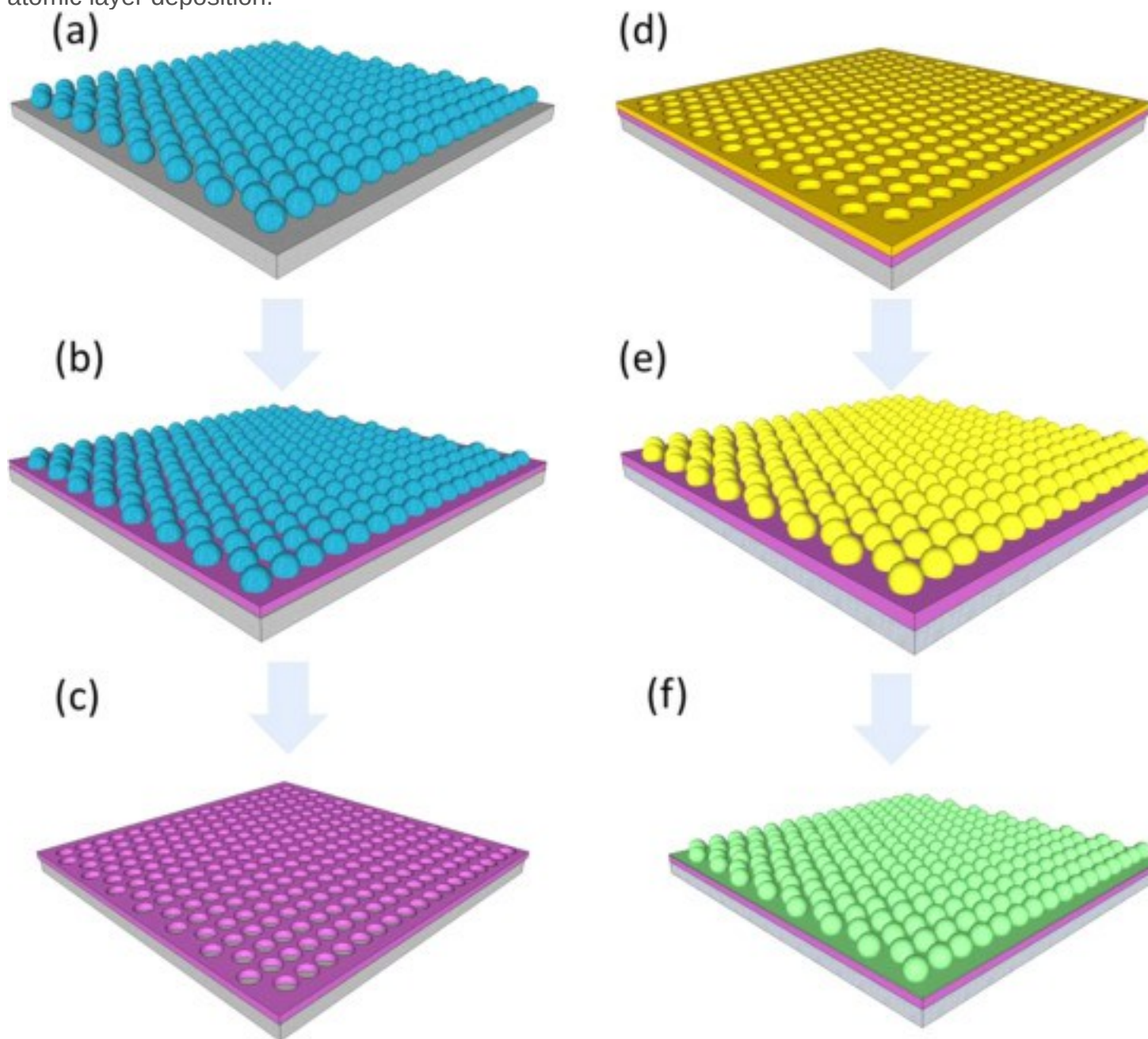
The normalized scattering spectra of nanocrystal arrays were measured by far-field [optical spectroscopy](#). A 100 W halogen lamp was used as the excitation source. The light from the source was focused and incident on the sample through a 100× (N.A.=0.95) objective lens by dark-field illumination. With the diameter of the collection region being 2 mm, the scattered light was collected by the same objective lens and was transmitted into a spectrometer (BTC 111E, B&W tec, USA) with an optical fiber. The obtained spectra were subjected to proper boxcar smoothing and intensity normalization, and the background signal was all subtracted.

2.3. Measurement of the Hydrogen production by gas chromatography

In the photocatalytic experiment, the Au/TiO₂ nanocrystal arrays (1 cm×1 cm) were utilized as the [photocatalysts](#) with various TiO₂ thickness or Au sizes in a 1:4 v/v methanol/aqueous solution. The solution and the catalysts were transferred into a cylindrical reactor connected to [gas chromatography](#) system (Shimadzu GC-2014, the whole volume is 65 ml, equipped with a gas inlet valve and a gas outlet valve.). Purged with Ar for 1 h, the reactor was tightly closed. The [irradiation](#) source was a 200 W Hg lamp as the UV light source, while a 150 W Xe lamp was utilized as the visible light source. Both lamps irradiated with a distance of 5 cm between the light source and the surface of the reactor.

3. Results and discussion

[Fig. 1](#) schematically illustrates the fabrication sequence of Au/TiO₂ hybrid [nanocrystal](#) arrays. The whole fabrication process can be divided into two steps: the fabrication of hexagonal Au nanocrystal arrays and the deposition of TiO₂ by atomic layer deposition.



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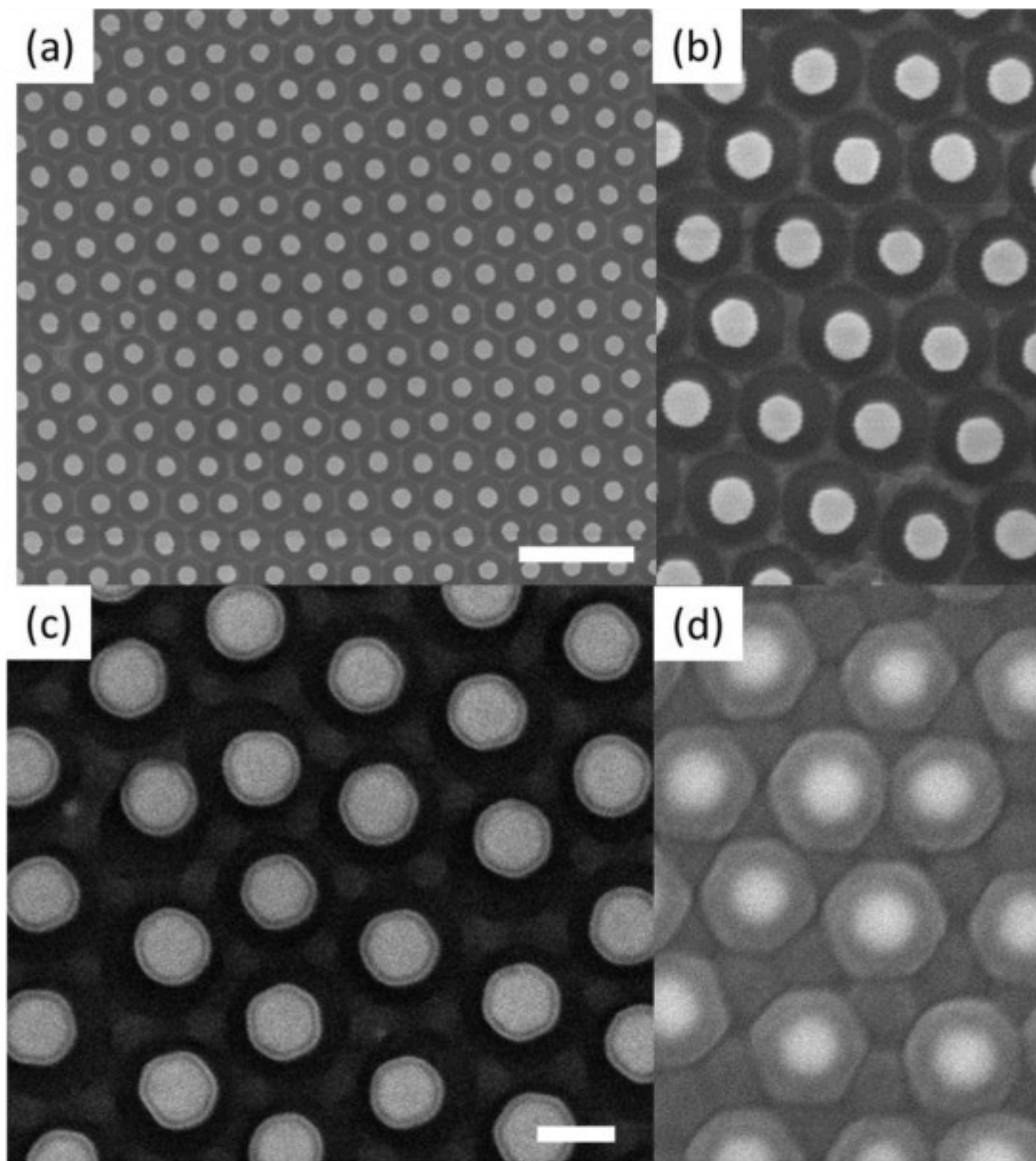
Fig. 1. Fabrication sequence diagrams of the hexagonal Au/TiO₂ hybrid [nanocrystal](#) arrays: (a) arrangement of monolayer [polystyrene](#) (PS)

spheres; (b) deposition of SiO₂ layer; (c) removal of the PS spheres; (d) deposition of Au film; (e) [annealing](#) to form the close-packed Au nanocrystal arrays; (f) deposition of TiO₂ by atomic layer deposition.

For the formation of Au nanocrystal arrays, the monolayer [polystyrene](#) spheres were first dispersed on a clean Si substrate as the mask to define the nanocrystal arrays. After O₂[plasma etching](#), the hexagonal SiO₂ layer was deposited by an electron beam evaporation system to form the honeycomb walls. Then the PS spheres were removed with the SiO₂ layer remained, followed by the deposition of the Au film. With SiO₂ network serving as separation ridges, the Au nanocrystal arrays were formed during the [annealing](#) process. ([Fig. 1a~e](#)) In our previous work, the Au nanocrystals were found to be single [crystalline](#), which is beneficial to LSPR properties for the internal losses derived by plasmons scattering at [grain boundaries](#) would be avoided and the maximal near-field intensity enhancement would be reached [\[2\]](#).

After the colloidal [lithography](#) and the annealing process, TiO₂ layer was deposited by atomic layer deposition (ALD). Atomic layer deposition is well-known for excellent step coverage, uniform deposited film and high precision [film thickness](#). Those advantages contributed to the [surface reaction](#) mechanism of [chemisorption](#) saturation and self-limiting film growth, differing from the traditional deposition technology. Furthermore, the TiO₂ layers coated by ALD were analyzed to be amorphous. (See the [Supporting Information, S1](#)) With the localized electronic states due to the structural disorder inside the bandgap, the amorphous TiO₂ are more likely to increase the absorption [\[17\]](#).

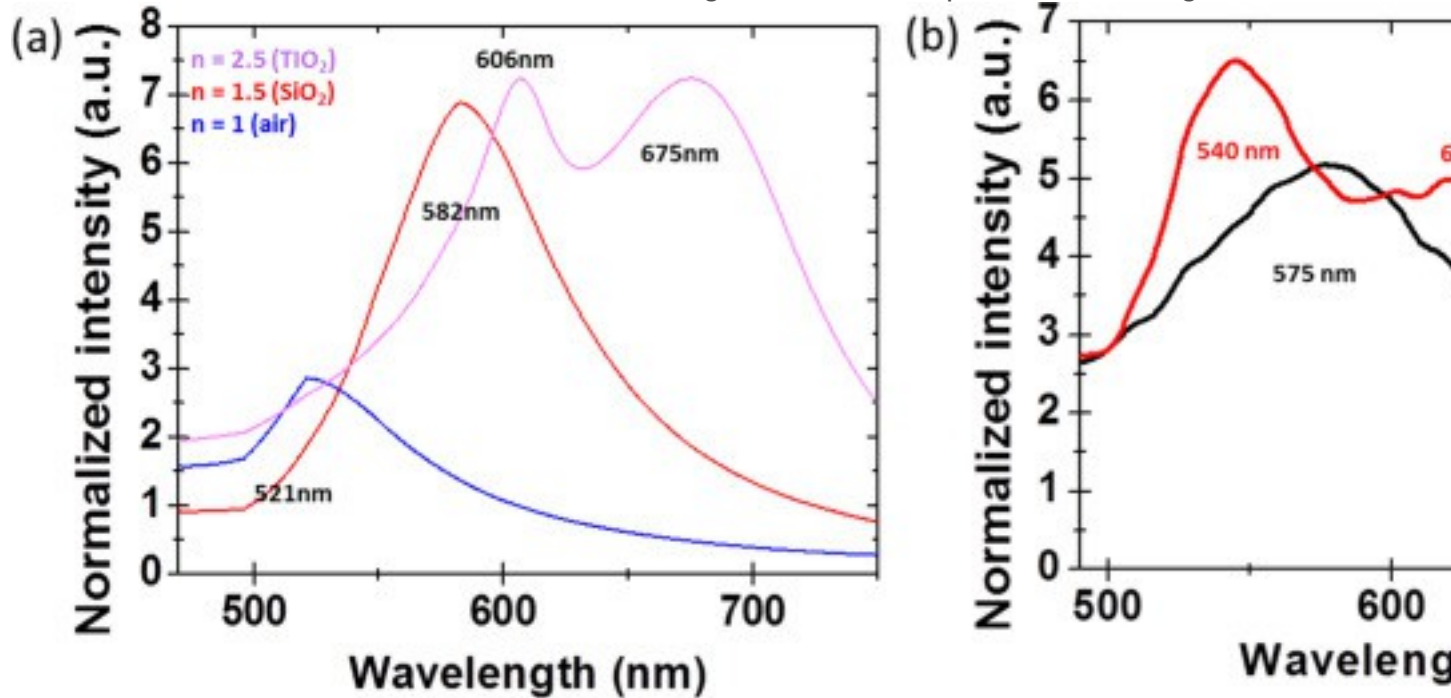
By separately introducing the precursors into the chamber, the TiO₂ film thickness can be precisely controlled. It is known that the LSPR wavelength and localized electric field enhancement are highly dependent on the surrounding media. The top view SEM images are shown in [Fig. 2](#). (Cross-section TEM image is shown in [S2](#)) [Fig. 2a](#) shows a low magnification SEM image revealing ordered periodicity of Au nanocrystal arrays over a relatively large area. In [Fig. 2b](#), it can be seen that the Au particles are located at the middle of the honeycomb SiO₂ walls. In [Figs. 2c](#) and [d](#), it is clear that the TiO₂ shells coated by ALD were uniformly distributed on the Au nanocrystal arrays, and the thickness of the TiO₂ shell is nearly the same. In addition, the periodicity of the Au nanocrystal arrays was maintained after the ALD deposition process.



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Fig. 2. Top view SEM images of the hexagonal close-packed [nanocrystal](#) arrays: (a) low magnification (with the scale bar corresponding to 500 nm) and (b) high magnification images; (c) 10 nm TiO₂/Au (shell/core) deposited by ALD; (d) 40 nm TiO₂/Au (shell/core) deposited by ALD. The scale bars in (b)-(d) correspond to 100 nm.

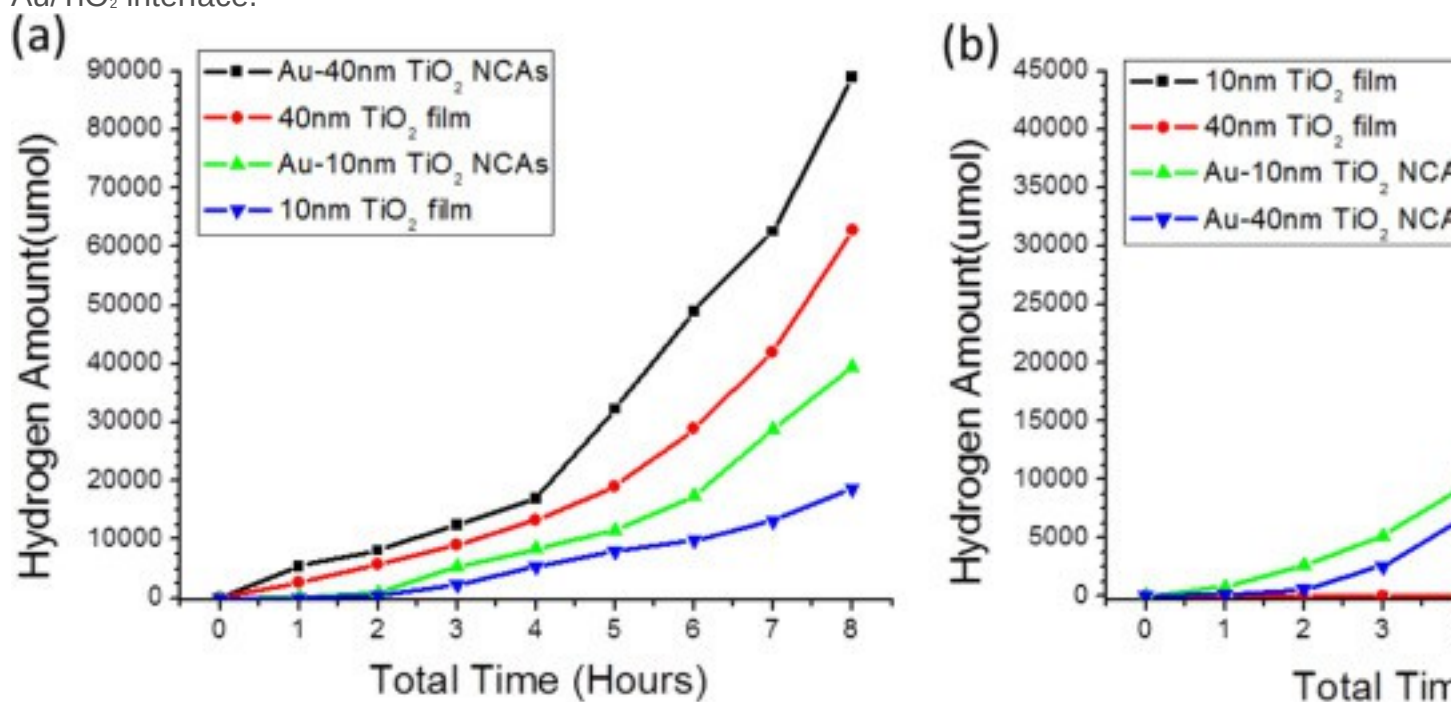
It is known that the wavelength and intensity of LSPR depend on the morphology of the [nanostructures](#) and local [dielectric](#) environment [1], [2]. Fig. 3a shows the simulation of scattering spectra of Au nanocrystal arrays with different surrounding media based on Mie's theory. The extinction, scattering and absorption efficiencies can all be considered in the simulations. To simplify the simulations, spherical nanocrystal was assumed and coupling effect was neglected. The normalized measured scattering spectra of nanocrystal arrays by far-field [optical spectroscopy](#) are shown in Fig. 3b. A 100 W halogen lamp was used as the excitation source. The detailed measurement method and simulation has been reported in our previous paper and are described in S3[2], [28]. The background signal had already been subtracted from the spectra. The LSPR wavelengths (λ_{LSPR}) are marked in the figure. These nanocrystal arrays exhibit λ_{LSPR} in the range of 500–650 nm. The λ_{LSPR} of bare Au nanocrystal arrays is about 575 nm, which agrees with that of our previous work [2]. After deposition of 10 nm TiO₂, two λ_{LSPR} peaks, 540 nm and 620 nm, were found. The splitting of the peaks can be explained by Mie theory since the Au in Au/TiO₂ nanocrystal arrays were only encircled partly by TiO₂ and partly by SiO₂. As a result, the scattering may result in two peaks located near 521 nm and between 606 nm and 675 nm, which are in agreement with experimental findings.



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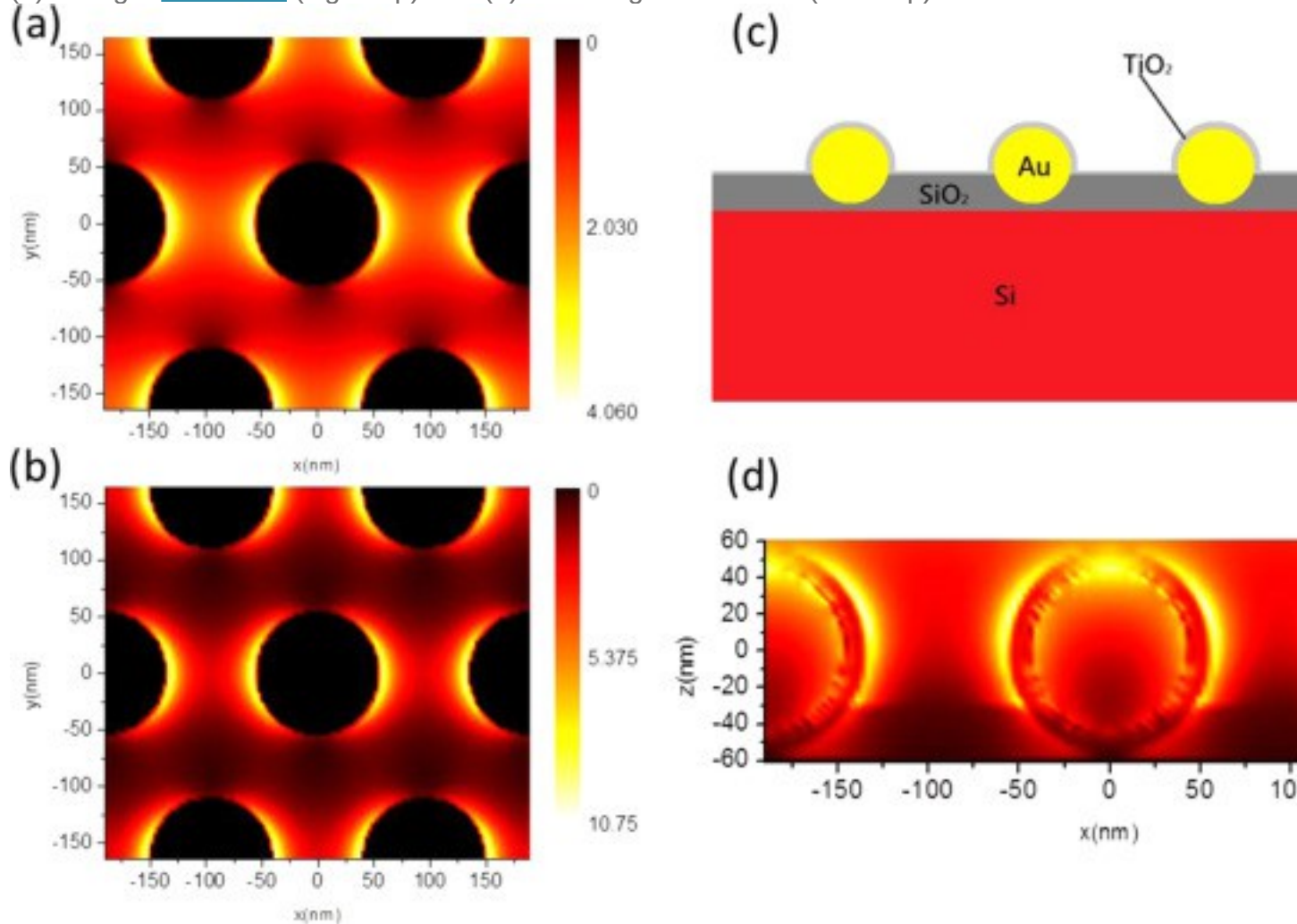
Fig. 3. (a) Simulated scattering spectra for Au [nanocrystal](#) arrays with crystal size of 100 nm in different surrounding media: TiO_2 , SiO_2 and air. (b) Measured scattering spectra for Au nanocrystal arrays with Au size of 100 nm and Au/ TiO_2 nanocrystal arrays with 10 nm TiO_2 . A 100 W halogen lamp was used as the excitation source.

The photocatalytic properties of the Au/ TiO_2 nanocrystal arrays were investigated with a 1:4 v/v methanol/aqueous solution. The hydrogen production results under UV-light [irradiation](#) are shown in [Fig. 4a](#). Compared to those with bare TiO_2 film, Au/ TiO_2 nanocrystal arrays display significantly higher photocatalytic activities. From [Table S1](#), for both 10 and 40 nm TiO_2 films, the hydrogen production rates increase substantially with the presence of Au/ TiO_2 NCAs. The occurrence of the plasmon enhanced electric field in Au particles improves the probability of the generation of the electron-hole pairs. In addition, the electrons generated in TiO_2 move to Au regions, which also act as catalytic sites, thus prolong the lifetime of the electron and hole pairs. In [Fig. 5a](#), FDTD simulation shows the near-field was increased by 3 times from the injected UV-light. Furthermore, although the hydrogen production rate increases as the TiO_2 thickness increases, the plasmon photocatalytic enhancement is more obvious for 10 nm TiO_2 (175%) than 40 nm TiO_2 (117%). It is likely due to exponential decay of the plasmon enhanced near-field in TiO_2 from the Au/ TiO_2 interface.



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Fig. 4. Amount of hydrogen generated with samples of different coating thickness of TiO_2 forming Au/ TiO_2 NCAs as [photocatalysts](#) compared with bare TiO_2 thin films under (a) UV-light [irradiation](#) (Hg lamp) and (b) visible-light irradiation (Xe lamp).



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Fig. 5. The mappings of electric near-fields simulated by [finite difference](#) time domain method for Au-10 nm TiO_2 [nanocrystal](#) arrays under different wavelengths polarized along x-axis, (a) 380 nm and (b) 620 nm. (c) Schematic diagram of cross-section of the Au/ TiO_2 nanocrystal arrays. (d) The mapping of cross-section near-fields for Au-10 nm TiO_2 nanocrystal arrays partially immersed in SiO_2 simulated by FDTD, $\lambda=540$ nm.

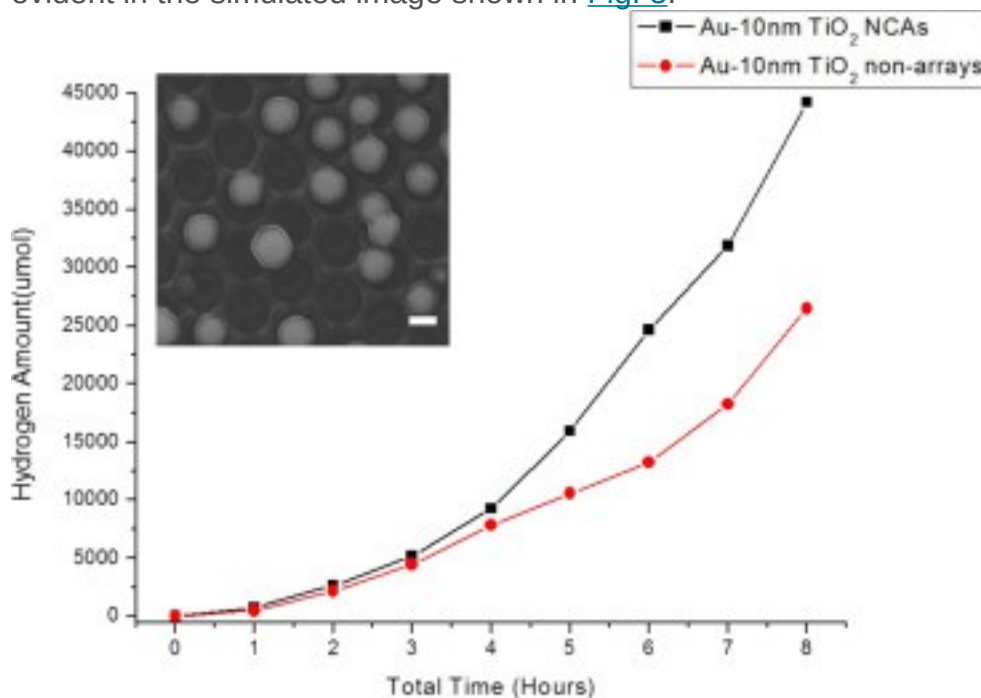
On the other hand, the photocatalytic activities of the bare TiO_2 thin films of different thicknesses (10 and 40 nm) were at very low level under visible-light irradiation (cutoff filter, $\lambda < 440$ nm) as shown in [Fig. 4b](#). It can be understood that the direct

photoexcitation of TiO_2 occurs only when the energy of the injecting photons is higher than the bandgap of TiO_2 , that is, $\lambda < 380 \text{ nm}$. Consequently, visible-light irradiation of the TiO_2 thin film results in very low level hydrogen production.

In contrast, for Au/TiO_2 NCAs, the visible-light irradiation can stimulate the generation of the electron-hole pairs in TiO_2 . [Fig. 4b](#) shows that the Au/TiO_2 NCAs induced high photocatalytic activities. Furthermore, the photocatalytic activity of Au/10 nm TiO_2 NCAs is higher than that of Au/40 nm TiO_2 NCAs. It is thought that, under visible-light irradiation, not only the excitation of the LSPR takes place, but the surface plasmon excitations under the enhanced plasmon near-fields close to Au/TiO_2 interface also results in the generation of [hot electrons](#). In [Fig. 5b](#), the near-field intensity is seen to increase by about 10 times. Moreover, with the corroboration of photocatalytic activity under illumination of visible-light being higher than that of UV-light, the plasmon-induced hot electrons are more efficient and have higher energies than those from direct photoexcitation [\[26\]](#), [\[27\]](#). As the electrons migrate to the TiO_2 conduction band, the probability of recombination of the electron-hole pairs decreases. Since the plasmonic near-fields are localized close to the Au/TiO_2 interface, the increase of the thickness of TiO_2 has little influence on the effect of near-fields intensity. As a result, the photocatalytic activity of Au/10 nm TiO_2 NCAs is higher than that of Au/40 nm TiO_2 NCAs mainly due to shorter distance for migration of electrons in thinner TiO_2 . Accordingly, in thicker TiO_2 layer, the recombination is more prone to occur; hence the photocatalytic activity is lower.

On the other hand, the surrounding dielectric media also play an important role in the enhancement of the plasmonic near-fields. From the cross-section structure of the Au/TiO_2 NCAs illustrated in [Fig. 5c](#), it can be seen that a non-symmetric structure encloses the Au [nanoparticles](#) with the upper and lower surroundings being TiO_2 and SiO_2 , respectively. The FDTD near-fields intensity simulation is shown in [Fig. 5d](#). It is obvious that the near-field intensity in TiO_2 is greater than in SiO_2 . It is the difference in dielectrics in surrounding media (SiO_2 refractive index=1.5, TiO_2 refractive index=2.5) that made the near-field intensity distribution asymmetric and concentrated in TiO_2 [\[17\]](#). To investigate whether the regular spacing of NCAs affected the efficiency of hydrogen production compared to the randomly distributed Au/TiO_2 particles, further photocatalytic measurements were carried out. The morphology of the randomly distributed Au NPs is shown in the inset of [Fig. 6](#). With significant variation in size of Au cores and different core-to-core spacings, the Au/TiO_2 nanocomposites exhibit no apparent periodicity. [Fig. 6](#) shows that for samples with the presence of Au/TiO_2 NCAs, higher photocatalytic activities occur than those with randomly distributed Au/TiO_2 nanocomposites.

From [Table S2](#), considering randomly distributed Au/TiO₂ core-shell structure, the Au part still enhances TiO₂ surface electric field due to LSPR, the lateral coupling between neighboring Au particles has been less efficient with varied core-to-core spacings. In contrast, for Au/TiO₂ NCAs, not only the electric field is enhanced at interfaces along all directions, but the lateral coupling also accentuated the enhancement significantly, as evident in the simulated image shown in [Fig. 5](#).



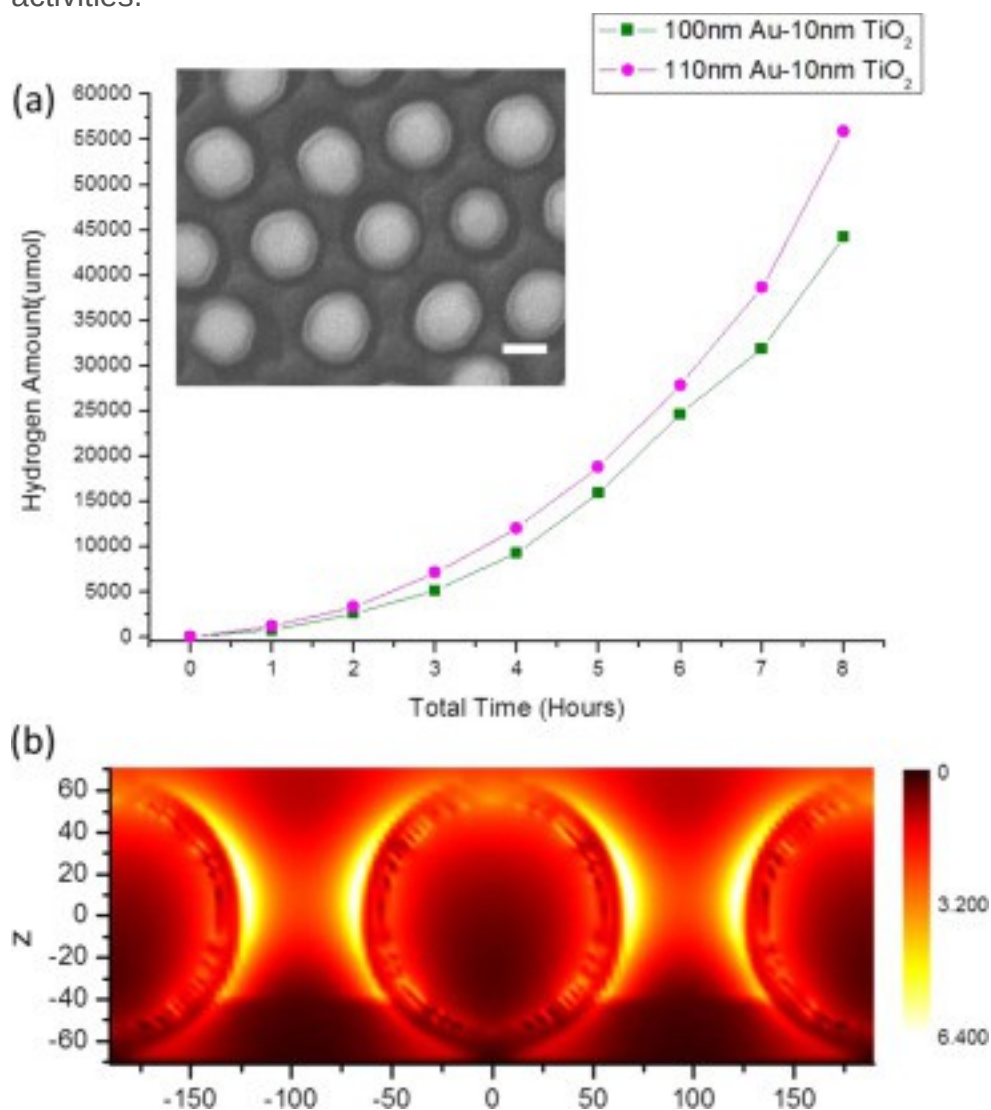
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Fig. 6. Amount of hydrogen produced from Au/TiO₂ NCAs and randomly distributed Au/TiO₂ [nanoparticles](#) with the same thickness of TiO₂ under visible light [irradiation](#). Inset is the SEM image of randomly distributed Au/TiO₂ nanoparticles. The scale bar corresponds to 100 nm.

Since the well-arranged core-to-core spacing of the Au/TiO₂ NCAs does influence the electric field distribution, enlarging the Au core with the same spacing may increase the plasmon-enhancement of the near-field.

[Fig. 7a](#) and [Table S3](#) show the amount of hydrogen generated from Au/TiO₂ NCAs with the same thickness of TiO₂ (10 nm) but different sizes of Au core (110 nm and 100 nm) under visible light irradiation. Inset is the SEM image of 110 nm Au core/TiO₂ NCAs. It is apparent that with the same TiO₂ thickness, the larger the Au core, the higher the H₂ production efficiency. [Fig. 7b](#) shows calculated cross-section near-field distribution with lateral coupling. The electric field intensity is enhanced up to 6.4 times, instead of

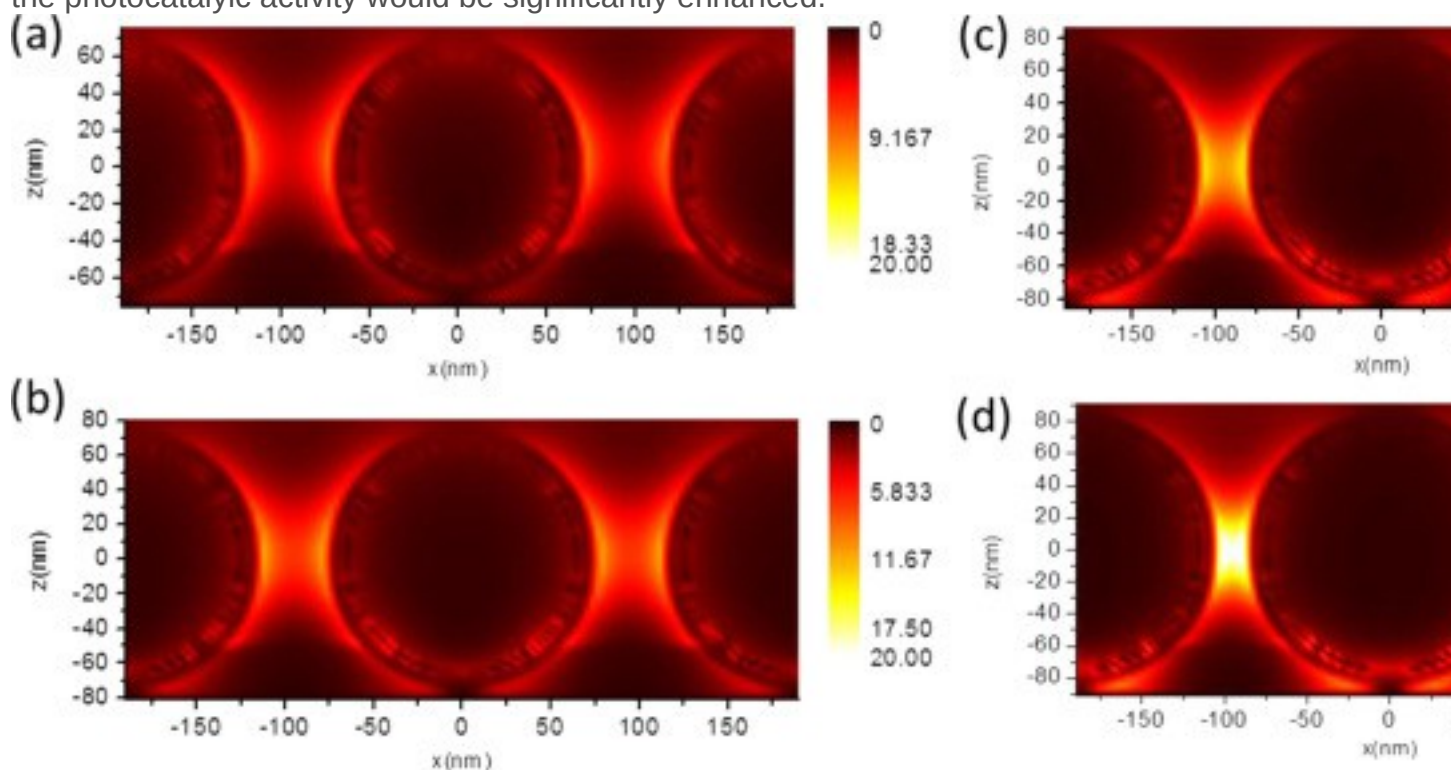
4.6 times for 100 nm Au/TiO₂ NCAs shown in [Fig. 5d](#). Accordingly, the photocatalytic efficiency of 110 nm Au/TiO₂NCAs increases by about 16.67%. The results indicated that with larger Au core, hence closer distance between Au nanoparticles, stronger coupling between neighboring nanoparticles led to higher plasmon-enhanced photocatalytic activities.



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Fig. 7. (a) Amount of hydrogen generated from Au/TiO₂ NCAs with the same thickness (10 nm) of TiO₂ but larger Au core (110 nm) under visible light [irradiation](#). Inset is the SEM image of 110 nm Au core /TiO₂ NCAs. The scale bar corresponds to 100 nm. (b) The mapping of cross-section near-fields simulated by FDTD method, $\lambda=550$ nm. Moreover, to explore further the plasmonic resonance behavior of the enlarged Au core, we have carried out a series of near field simulations by FDTD. The mapping of cross-

section near field with the size of the Au core being 120, 130, 140 and 150 nm and with peaks of the scattered spectra $\lambda=560, 570, 580$ and 590 nm, respectively, indicates the red shift of LSPR peaks with the enlarged Au core [2]. As shown in Fig. 8, the maximum intensity of the near field was increased from 9.2 times for the 120 nm Au cores to 18.9 times for the 150 nm Au cores, indicating that the lateral coupling was enhanced remarkably as the size of Au core increased. Therefore, with a greater lateral coupling, the photocatalytic activity would be significantly enhanced.



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Fig. 8. The mappings of cross-section near-fields simulated by FDTD method, the size of the Au core are (a) 120, (b) 130, (c) 140 and (d) 150 nm, with peaks of the scattered spectra $\lambda=560, 570, 580$ and 590 nm, respectively. To emphasize the enhanced near-field intensity, the contrast of the color scale was adjusted to be in the same range, with the maximum intensity being 9.2, 11.75, 14.4, 18.9, respectively.

[Table S4](#) lists the comparison of present results with reported data in the literatures for Au-TiO₂ system and some other catalysts [29], [30], [31], [32], [33], [34], [35], [36].

Although the hydrogen production rate may vary significantly with the water/methanol ratio, intensity of the irradiation source, experimental set-up etc. The photocatalytic performance of Au-10 nm TiO₂ NCAs stands out as remarkably high. The results

indicate that the plasmonic photocatalytic enhancement is very effective with the well-designed metal/semiconductor hybrid structure.

4. Conclusions

In conclusion, we have successfully fabricated the hybrid Au/TiO₂ [nanocrystal](#) arrays with highly ordered periodicity and uniformity. Under both ultra-violet and visible light, significant increase in the hydrogen production from methanol (20%) solution splitting was achieved with the hybrid Au/TiO₂ nanocrystal arrays in comparison with bare TiO₂ thin film as well as randomly distributed Au/TiO₂ nanocrystals. With larger [refractive index](#) TiO₂ as the surrounding [dielectrics](#), the LSPR of the Au/TiO₂ nanocrystal arrays red-shifts. With the strong localized plasmonic near-field enhancement at the Au/TiO₂ interface, the photocatalytic efficiencies increase under both UV-light and visible-light. On the other hand as the thickness of TiO₂ increased from 10 to 40 nm, the photocatalytic efficiency under visible-light [irradiation](#) decreased instead for the carrier recombination is more prone to occur in thicker TiO₂ layer. From the data of photocatalytic activities, the plasmon-induced [hot electrons](#) were found to be more efficient, compared to those from direct photoexcitation.

Furthermore, the lateral coupling has been systematically investigated for the first time. Owing to the highly ordered periodicity, the lateral coupling of the electric field is shown to be more significant compared to the randomly distributed Au/TiO₂ nanocrystals, hence higher photocatalytic activity, as obtained from both [theoretical calculations](#) and experimental findings. The Au core size and the TiO₂ shell thickness are also shown to affect the photocatalytic activities. The results are particularly useful in designing of optimal structures of regular metal-dielectric nanocrystals with appropriate spacing of nanocrystals, metal core size and dielectric shell thickness.

Notes

The authors declare no competing financial interests.

Acknowledgments

We are grateful to late Prof. Li-Jen Chou for helpful discussion. The research was sponsored by the Ministry of Science and Technology of Taiwan under grant no. MOST [104-2221-E-007-30-MY3](#).

Appendix A. Supplementary material

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Supplementary material **Supporting Information**. Characterization of ALD-deposited TiO₂, cross-section TEM images and Fig. S1–S4 are available in the supporting information. This material is available free of charge via the Internet at

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